

We built a reconciliation data platform — then used it to test how AI should review code

A real client needed reconciliation data made legible. We shipped a production platform for the **Indigenomics Institute** — and its own **30 merged pull requests** became the industrial dataset for a pre-registered study of multi-agent AI code review.

Research companion: “Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study”

En-Ping Su · Tong Wu · Shiting Huang · Mengshan Li

Khoury College of Computer Sciences, Northeastern University · Vancouver · Advisor: Yvonne Coady · Industry Partner: Indigenomics Institute (Carol Anne Hilton, Founder)

RESEARCH CONTRIBUTION

More AI reviewers don't help — **decorrelation does**. Agreement predicts truth only when the reviewers are independent.

89%

3 independent families agree → matches ground truth

54%

one model, three runs → mostly noise

01 THE PROJECT — RAP DATA PORTAL

PROBLEM

Companies publish reconciliation commitments. Legal precedents sit scattered across court records. There is no searchable, verifiable, unified view — so nobody can track progress.

OUR SOLUTION

▪Searchable RAP commitments

▪Legal precedent database

▪AI extraction from PDFs

▪Human review workflow

▪AWS serverless hosting

▪Authentication & roles

▪Progress tracking

▪Coverage analytics

WHO USES IT

▪Indigenomics Institute

▪Indigenous organizations

▪Corporate partners

▪Researchers & legal advisors

indigenomics · RAP Index

INDIGENOMICS

RAP Index

Organizations

Suppliers

Coverage analysis

Verification

Alignment

Extract

Notifications

Table

Explore

The RAP Index

commitments by sector, size & type

Reconciliation commitments across the network, and how they progress over time. Distinct from Spend Coverage, which tracks confirmed dollars.

Key takeaways

Plain-language highlights, generated automatically from the data below. Start here.

106 commitments across 100 organizations, averaging 63% progress. 0% already confirmed.

Procurement is the most common commitment type (36), while Employment lags on delivery at 53% average progress.

Of 29 claimed outcomes, 0% are supplier-confirmed. 29 remain self-reported and unverified.

Needs attention: 31 are past their target year without confirmation, and 43 due in 2026 are behind pace.

At a glance

The headline totals for the current view: commitments, organizations, average progress, and the confirmed share.

106

commitments

100

organizations

63%

average progress

0%

confirmed

indigenomics · Legal Cases

Legal Cases

Cases

Find similar

Activation

Legal info

Monitoring

Methodology

Unofficial reproductions of public court decisions · not legal advice · every claim links to its source

Legal cases — economic justice

Cenaka's Indigenous economic justice case law, searchable and citation-anchored.

Describe your situation to find similar cases →

View as

Indigenous government

Legal advisor

Corporate / advisory

Presidents affirming your community's economic rights and self-determination. The same public record, rendered for your context — anyone can switch, nothing is hidden.

Search citation, case name, or full text...

Search

Tier: auto

All themes

All courts

All outcomes

All nations

to yr

clear

from yr

563 results · showing 1-10 · browse · filter cases

Legal Cases

◀ 561 economic-justice decisions

◀ citation-anchored, role-based views

RAP Index

- ◀ 106 commitments · 100 organizations
- ◀ 63% average progress tracked

ARCHITECTURE — SERVERLESS ON AWS

CloudFront

edge

→

Next.js

on Lambda

→

Texttract

OCR

→

Bedrock

extraction

→

DynamoDB

x6 tables

WHAT THE INSTITUTE CAN NOW DO

▪See every tracked commitment and its progress in one view

▪Get a weekly digest of what has gone **overdue**

▪Trace any claim back to the published source document

▪Read the legal precedent beside the corporate promise

02 HOW IT BECAME RESEARCH

Client problem

reconciliation data scattered & unverifiable

↓

We built the portal

production platform, shipped on AWS

↓

179 pull requests

30 of them ours — real, merged, AI-reviewed

↓

Benchmark dataset

E1 Qodo · E2 SWE-PRBench · E3 our portal — 2,415 reviews

↓

Four AI review architectures

each rung adds exactly one capability

FOUR AI REVIEW ARCHITECTURES — EACH RUNG ADDS ONE CAPABILITY

Held constant: Same PR snapshot · Same model (Claude Haiku 4.5) · Same prompts · Three runs per arm · Pre-registered on OSF

(a) Agentless — Single LLM Baseline

+ Sampling

Pull Request (Diff)

Single LLM (One Pass)

Review Report

1 LLM Call

1 Message

(b) Generalists-3 — Compute-Matched Control

+ Specialization

Pull Request (Diff)

Generalist A

Generalist B

Generalist C

Deterministic Merge (No Communication)

Review Report

3 LLM Calls

3 Messages

(c) Hierarchical — Specialized Coordination

+ Communication (Manager)

Pull Request (Diff)

Coordinator (Deterministic Routing)

Frontend Specialist

Backend Specialist

Database Specialist

Synthesis & Aggregation (Deterministic)

Review Report

3 LLM Calls

8 Messages

(d) Consensus — Peer Collaboration

+ Communication (Peers)

Pull Request (Diff)

Protocol Scheduler

Frontend Specialist

Backend Specialist

Database Specialist

Exchange Findings (Challenge & Clarify)

Vote & Resolution (Consensus)

Review Report

9 LLM Calls

22 Messages

03 WHAT WE FOUND · E1 · QODO, 99 PRS

Structure is null. No multi-agent arm beats single-pass F1; specialists tie compute-matched sampling (p=0.73); peer debate adds nothing over a manager (p=0.12).

Architecture	F1 Score
Agentless (1 call)	0.487
Generalists-3 (3 calls)	0.357
Hierarchical (3 calls)	0.378
Consensus (9 calls)	0.369

Extra agents buy recall at a price — up to **5×** the cost per confirmed finding.

Architecture	F1 Score
Agentless (1 call)	0.34
Generalists-3 (3 calls)	0.45
Hierarchical (3 calls)	0.50
Consensus (9 calls)	1.76

Bounded. Recall plateaus at **≈0.83**; on our own unlabeled code, LLM judges disagree (**κ=0.03**).

WHERE THE SIGNAL LIVES

Agreement recovers **universal functional bugs** far better than project-specific conventions — which is why the ceiling exists.

43%

functional-bug recall at 3 independent families agreeing

18%

rule-violation recall at the same depth

Pair decorrelated review with a deterministic linter.

PROJECT IMPACT

▪Production platform delivered to the Indigenomics Institute

▪Reconciliation commitments made searchable and verifiable

▪Indigenous legal precedent explorer, citation-anchored

RESEARCH IMPACT

▪First compute-matched comparison of review architectures

▪Independence beats more agents — decorrelate your verifiers

▪Guidance for anyone deploying AI code review in production

Code & data
rap-review-research

Live portal
try the demo

Acknowledgements. Lino Coria Mendoza (weekly progress check-ins) · Indigenomics Institute: Shawn Anderson (technical contact), Eve Marengi (scheduling), K. Krasnow Waterman (legal advice) · Compute: AWS Bedrock.

Pre-registered on OSF (embargoed)